

MIDTERM GROUP PROJECT

WAREHOUSE INVENTORY & ORDER MANAGEMENT SYSTEM

I. Project Overview

For the Midterm Group Project, your group will design the User Interface (UI) and User Experience (UX) of a Warehouse Inventory & Order Management System.

The purpose of this project is to demonstrate your ability to design a system that is:

- Usable
- Understandable
- Consistent
- Organized
- Easy to navigate
- Efficient to use
- Able to prevent common user errors

The Midterm focuses on how the system should be designed before it is implemented.

You are not required to develop a backend or fully functional system for the Midterm.

II. System to Be Designed

Your group will design a:

Warehouse Inventory & Order Management System

The system will be used by warehouse personnel to manage products, inventory, deliveries, orders, stock movements, reports, and notifications.

The system must contain the following 7 modules:

1. Login & User Management
2. Warehouse Dashboard
3. Inventory Management
4. Receiving Management
5. Order/Picking Management
6. Stock Movement & Tracking
7. Reports & Notifications

III. Module Requirements

Module 1 — Login & User Management

The design must allow the user to:

- Enter username/email and password
- Log in
- Receive feedback when login information is incorrect
- Access the system after successful login
- View basic user information
- Log out

Expected Design

The interface should clearly communicate:

- Where the user should enter information
- Which fields are required
- How to submit the login
- What happens when incorrect information is entered
- How the user exits the system

Module 2 — Warehouse Dashboard

The dashboard must provide an overview of the warehouse.

It should display information such as:

- Total products
- Available stock
- Low-stock products
- Incoming deliveries
- Pending orders
- Recent activities
- Notifications or alerts

Expected Design

The user should be able to:

- Understand the current warehouse status quickly
- Identify important information
- Navigate to relevant modules
- Recognize warnings or important conditions
- Access frequently used functions

Module 3 — Inventory Management

The system must allow users to:

- View products
- Search products
- Filter products
- View product details
- Add products
- Edit products
- Delete products
- View stock quantity
- View warehouse location
- View stock status

Product information should include appropriate details such as:

- Product ID/SKU
- Product name
- Category
- Quantity
- Location
- Stock status
- Supplier

Expected Design

The interface should make it easy for users to:

- Locate a product
- Understand its current stock
- Add or modify product information
- Identify available actions
- Avoid entering invalid information
- Confirm important actions

Module 4 — Receiving Management

The system must allow warehouse personnel to record incoming goods.

The interface should include appropriate information such as:

- Supplier
- Delivery number
- Product
- Quantity
- Date
- Condition

The user should be able to:

- Enter receiving information
- Review the information
- Confirm the transaction

- Cancel the transaction
- Correct incorrect information

Expected Design

The interface should provide appropriate feedback and confirmation during the receiving process.

The user should be able to understand:

What is being received → How much is being received → From whom → When → What happens after confirmation

Module 5 — Order/Picking Management

The system must allow users to:

- View pending orders
- View order details
- View required products
- View required quantities
- Start the picking process
- Mark items as picked
- Update order status
- Complete an order

The system should clearly communicate order status, such as:

Pending → Picking → Ready → Completed

Expected Design

The interface should help users:

- Identify which order needs attention
- Understand what products need to be picked
- Know which items have already been picked
- Identify unavailable or insufficient stock
- Understand the current order status
- Complete the picking process correctly

Module 6 — Stock Movement & Tracking

The system must allow users to view inventory movement.

The interface should display information such as:

- Product
- Quantity
- Movement type
- Date/time
- Previous stock
- Updated stock

- Responsible user/action

Movement types may include:

- Received
- Released
- Transferred
- Adjusted

Expected Design

Users should be able to:

- Review stock movement history
- Search or filter records
- Identify increases and decreases in stock
- Understand how the current stock was affected
- Trace previous inventory activity

Module 7 — Reports & Notifications

The system must provide relevant warehouse notifications and reports.

Notifications may include:

- Low-stock alerts
- Pending orders
- New deliveries
- Completed orders
- Stock adjustments

Reports may include:

- Inventory report
- Low-stock report
- Stock movement report
- Order report

Expected Design

Users should be able to:

- Identify important notifications
- Distinguish important or unread notifications
- Access relevant information
- Search or filter reports
- Understand information presented in tables, summaries, or charts

IV. Required Screens

Each group must create at least one primary screen for each of the 7 modules.

Therefore, the minimum is:

7 primary module screens

However, some modules will naturally require additional screens to demonstrate their complete interaction.

For example:

- Product details
- Add/Edit Product
- Confirmation dialog
- Error message
- Success message
- Order details

These additional screens or states are allowed and expected when necessary.

Important

The requirement is not "create 10–12 screens."

The requirement is:

Create enough screens and interface states to demonstrate the complete functionality and user flow of the 7 required modules.

V. Shneiderman's Eight Golden Rules

The entire Midterm design must apply Shneiderman's Eight Golden Rules of Interface Design:

1. Strive for consistency
2. Seek universal usability
3. Offer informative feedback
4. Design dialogs to yield closure
5. Prevent errors
6. Permit easy reversal of actions
7. Keep users in control
8. Reduce short-term memory load

The group must ensure that these principles are reflected throughout the design of all seven modules.

VI. Required User Flows

Your design must demonstrate at least three complete user flows.

User Flow 1 — Receive New Inventory

Login → Dashboard → Receiving → Enter Information → Review → Confirm

User Flow 2 — Process an Order

Dashboard → Orders → Select Order → Picking → Update Status → Complete

User Flow 3 — Manage Low Stock

Dashboard → Low Stock → Inventory → Product Details → Review

The user flow should be logical and understandable.

VII. Midterm Acceptance Criteria

Your Midterm project will be considered complete when:

General System

- The system has all **7 required modules**.
- Each module has at least one primary UI screen.
- Additional screens/states are provided where necessary to demonstrate interactions.
- The system has a clear navigation structure.
- The interface uses consistent visual and interaction patterns.
- The design supports the required user flows.
- The interface demonstrates Shneiderman's Eight Golden Rules.
- The design provides appropriate error, confirmation, and feedback states where necessary.
- The UI is understandable without requiring the presenter to explain every element.

Module Requirements

- Login & User Management meets its required functions.
- Dashboard provides a clear warehouse overview.
- Inventory Management provides appropriate product and stock controls.
- Receiving Management supports the complete receiving process.
- Order/Picking Management supports the order-picking process.
- Stock Movement & Tracking provides understandable inventory history.
- Reports & Notifications provide meaningful warehouse information.

User Experience

- Users can determine what actions are available.
- Users can understand the result of important actions.

- Users can identify and correct errors.
- Navigation is predictable.
- Users can cancel appropriate actions.
- Important information is easy to find.
- The design minimizes unnecessary memory load.

VIII. Midterm Deliverables

Each group must submit:

9. System Description
10. Target Users
11. System Requirements
12. Seven Modules
13. Sitemap
14. User Flow
15. UI/UX Design
16. Interactive/Clickable Prototype
17. Application of Shneiderman's Eight Golden Rules
18. Design Rationale

IX. Midterm Presentation

Each group must present:

19. System overview
20. Target users
21. Seven modules
22. Sitemap
23. User flows
24. UI/UX design
25. Demonstration of the prototype
26. Application of Shneiderman's Eight Golden Rules
27. Design decisions and rationale

The group must be able to explain why the interface was designed in a particular way, not simply show the screens.

X. Midterm Objective

At the end of the Midterm, your group should have a complete and approved UI/UX design that can be developed further for the Final Project.